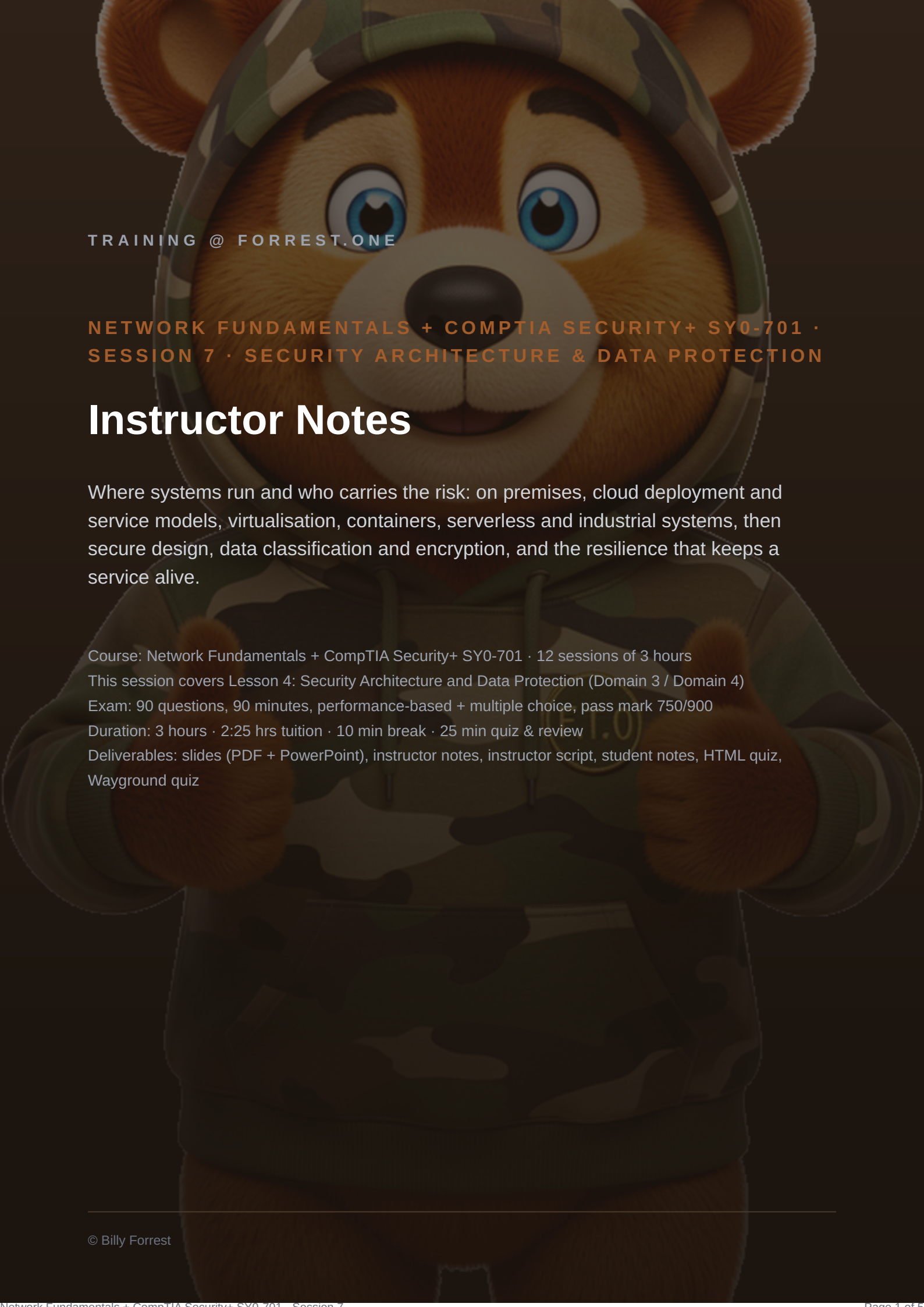




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 7 · SECURITY ARCHITECTURE & DATA PROTECTION

Instructor Notes

Where systems run and who carries the risk: on premises, cloud deployment and service models, virtualisation, containers, serverless and industrial systems, then secure design, data classification and encryption, and the resilience that keeps a service alive.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session covers Lesson 4: Security Architecture and Data Protection (Domain 3 / Domain 4)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Notes — Security Architecture & Data Protection

A 3-hour delivery guide for Session 7. This is the design session: the decisions here last for years, and the exam tests them through scenarios. Keep the activities focused on finding the responsibility line and choosing the right control for a data state, because that is exactly what the paper rewards.

Session at a glance

Time	Block	Slides	Focus / activity
0:00–0:10	Welcome and objectives	1–3	Where this session sits; the six-chapter shape
0:10–0:42	Architecture models	4–7	On premises; deployment models; service models; responsibility activity
0:42–1:12	Compute and edge platforms	8–14	Hypervisor; containers; serverless; IoT; SCADA; IT and OT discussion
1:12–1:38	Secure design	15–20	Infrastructure as code; network design; the channel; control selection
1:38–2:02	Data protection	21–27	Types and states; classification; at rest; disposal; in transit
2:02–2:15	Resilience and recovery	28–33	Availability; sites; backup and objectives; power; testing
2:15–2:25	Break	34	10-minute break, then the continuity chapter
2:25–2:35	Continuity and consolidation	35–36	Business continuity planning; business impact analysis
2:35–3:00	Exam map, quiz and close	37–41	Objective map; Quiz_Session_07.html; takeaways; preview

Materials and preparation

- Projector or screen with the Session 7 slide deck (02_Slides.pdf or 02_Slides.pptx).
- One computer or tablet per pair for Quiz_Session_07.html.
- Printed or digital copy of the Student Notes for each candidate.

- Whiteboard or flipchart: draw the responsibility line for one cloud service as a class.
- Optional: a vendor shared responsibility diagram, to compare with the one you draw.

Delivery guidance by slide

Slides 1–3 · Welcome (10 minutes)

Frame the session as the design layer. Session 6 gave the mechanisms; today decides where they are applied. Emphasise that architecture decisions are long lived, so a mistake here persists for years. Walk the six-chapter shape so candidates know where they are going.

Slides 4–7 · Architecture models (32 minutes)

On premises is about control and its price: patching, maintenance, disaster recovery planning and skilled staff all stay with you. The deployment models table (slide 5) is the backbone: each model shifts the security duties between provider and customer. For service models (slide 6), teach the pattern rather than the acronyms. The activity (slide 7) is the fastest way to make it stick.

Activity (slide 7)

Read the responsibility line in three cases: a virtual machine you patch yourself (infrastructure as a service), application code on a managed runtime (platform as a service), and a hosted mail suite where you manage permissions (software as a service). 6 minutes in pairs, 4 minutes debrief on what the customer still owns in each.

Slides 8–14 · Compute and edge platforms (30 minutes)

Virtualisation: everything depends on the hypervisor, so it is patched first and monitored hardest. Type 1 runs on hardware with a smaller attack surface; type 2 runs on a host operating system, so a flaw there reaches every guest. Virtual machine escape is the attack to name, and sprawl is the quieter risk. Containers are the same idea with a smaller boundary, and the image is the risk, because a vulnerable library is reproduced in every copy. Serverless trades control for less to manage, and the trigger becomes the entry point. For IoT, teach the four risks: too small, rarely updated, default credentials, and enormous scale. For ICS and SCADA, lead with the human consequence, because safety and availability outrank confidentiality. Close the block with the IT and OT discussion (slide 14).

Discussion (slide 14)

IT protects confidentiality; OT protects safety and availability. When the networks are joined for efficiency, an intruder who reaches the office can reach the plant. Steer the group to deliberate segmentation, industrial-aware monitoring, and testing before reliance, rather than refusing convergence outright.

Slides 15–20 · Secure design (26 minutes)

Infrastructure as code is the headline: the same file builds the same server every time, which removes drift. Then teach the matching risk immediately, because one misconfiguration in a file is reproduced everywhere

that file is used, and secrets written into files leak through repositories. Secure network design (slide 18) is the containment lesson: defence in depth, segmentation with virtual local area networks, access control lists, a demilitarised zone for public facing services, and least connectivity. Slide 19 covers the channel, including jump servers for administrators. Control selection (slide 20) is a decision, not a shopping list: risk first, then compliance, cost and operational impact, with technical, administrative and physical controls working together.

Slides 21–27 · Data protection (24 minutes)

Start with the legal weight of data types (personally identifiable information, protected health information, financial data) and then the three states. Classification (slide 23) is the control that makes handling possible; slide 24 shows how the label changes storage, transmission and disposal. For data at rest (slide 25), full disk encryption is the laptop answer and key management is the real subject, including hardware security modules. Disposal (slide 26) is where good intentions leak: wiping works on both hard disk and solid state media, degaussing destroys magnetic media only, and destruction is the strongest option. Data in transit (slide 27) is about using the secure version of every protocol, data loss prevention for outbound content, and validating certificates rather than clicking past warnings.

Slides 28–36 · Resilience and continuity (23 minutes, including the break)

The break (slide 34) closes this block: take it after slide 33, then return for the continuity chapter. Availability (slide 29) is measured and designed: uptime, mean time between failures, RAID, teamed cards, clusters, load balancing, failover, and the removal of single points of failure. Alternate sites (slide 30) is a cost against readiness decision, with geography as the balancing act. Slide 31 pairs the backup strategies with the recovery objectives, and the rule to teach is that the objective comes first. Slide 32 covers power, environment and platform diversity. Testing (slide 33) makes the point that an untested plan is only a guess. Business continuity (slide 36) is the umbrella: business impact analysis, acceptable downtime, continuity of operations, IT disaster recovery, and a plan that is maintained and rehearsed.

Slides 37–41 · Exam map, quiz and close (20 minutes)

Walk the objective map (slide 37) so candidates see the coverage. Run the quiz (slide 38) and review the weakest areas. Deliver the takeaways (slide 39), set the revision checklist (slide 40), and preview Session 8, which moves from architecture to network security controls.

Common misconceptions and corrections

Misconception	Correction
"Moving to the cloud moves the responsibility away"	It moves the responsibility, it does not remove it. Find the line for the service model you use
"Containers are secure because they are small"	The boundary is smaller, not absent. A vulnerable image reproduces its weakness in every copy
"Serverless means no security work"	The trigger is an entry point. Authenticate and validate every event that invokes a function

Misconception	Correction
"Industrial systems are safe because they are isolated"	OT and IT converge for efficiency. An intruder in the office network can reach the plant
"Infrastructure as code is automatically safer"	It gives consistency and misconfiguration at scale. Secrets in the file leak through repositories
"Encryption at rest solves stolen media"	Only if key management holds. Encryption is as strong as the handling behind the key
"Degaussing works on any drive"	It destroys magnetic media only. Solid state drives need wiping or destruction
"Backups are a configuration choice"	The recovery point objective comes first, and the backup design follows it
"A written plan is a plan"	An untested plan is only a guess. Rehearse it with the people who will use it

Exam alignment check

- Architecture modelling, the shared responsibility line and control selection are all directly examinable.
- Data classification, data states and encryption in each state appear across multiple domains.
- Resilience vocabulary (recovery point objective, recovery time objective, hot, warm and cold sites) is heavily tested.
- Quiz tags map every question to its exam objective for traceable revision.